

Cracking some number theory nuts with a sledgehammer

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This lecture explores how powerful mathematical tools—our “sledgehammers”—can be used to tackle seemingly simple yet stubborn problems in number theory. By showcasing these methods, the talk highlights both the elegance and necessity of deep mathematical machinery in solving hard problems. Whether you’re familiar with number theory or just curious about mathematical problem-solving at its finest, this lecture will offer a compelling perspective on when and why we need a heavy-duty approach. Problems 1–7 are intended for warming up; they are mostly one-liners, sometimes can even be interpreted as jokes (or, if you want, trolls). Among the rest of the problems, there are still two or three one-liners, but there are also some tougher challenges. Their order, however, does not unravel anything about their difficulty; in fact, the order is alphabetical.

1. Let n be a positive integer, $n \geq 3$. Prove that the number $\sqrt[n]{2}$ is irrational.
2. Let m and n be perfect 17^{th} powers with 15 062 017 and 15 062 017 000 digits, respectively. Can the number obtained by writing m and n one after the other be a perfect 17^{th} power?
3. Does there exist a 3×3 magic square all of whose entries are distinct prime numbers?
4. A finite sequence of consecutive positive integers contains at least one prime number. Prove that the sequence contains a number that is coprime to all the other terms of the sequence.
5. Do there exist positive integers n_1 and n_2 such that $n_1! \cdot n_2!$ is a perfect square and $|n_1 - n_2| \geq 2$?
6. Let $\pi(n)$ denote the number of prime numbers less than or equal to n . Prove that there exist infinitely many positive integers n such that $\pi(n) \mid n$.
7. Find all integer solutions (x, y, z) of the equation

$$(x + y + z)^5 = 80xyz(x^2 + y^2 + z^2).$$

8. A number of red, blue, and white baskets are given (there is at least one basket of each color). Any two baskets are different. There are n different pencils to be distributed to these baskets. Alice counted the number of such distributions under the constraint that there is at least one non-empty red basket and at least one non-empty blue basket. Bob counted the number of such distributions under the constraint that all the red and all the blue baskets are empty. It turned out that Alice and Bob obtained the same result. Assuming that there are exactly 2020 baskets of the most common color, what is the total number of baskets?
9. A positive integer n is called an *IMSC number in base b* if it can be obtained by applying the four basic operations, exponentiation, and concatenation, possibly with parentheses, to its own digits in the base b , where each digit is used exactly as many times as it appears in n , and the digits must appear in the same order as they do in n . (For example, 127 is an MBL number in base 10, since $127 = -1 + 2^7$.)
 - a) Find all bases b in which there are infinitely many IMSC numbers.
 - b) Find all bases b in which there are infinitely many prime IMSC numbers.(Clarification: at least one operation that is not concatenation must be applied.)
10. A positive integer n greater than 3 is given. Prove that we can find positive integers a, b, c, d (not necessarily distinct) such that

$$n = \varphi(a) + \varphi(b) + \varphi(c) + \varphi(d).$$

11. Assume that the twin prime conjecture is true (that is, there are infinitely many prime numbers that differ by 2).
- Prove that there exist infinitely many pairs of positive integers m and n such that $\sigma(m) = \varphi(n)$.
 - Prove that there exist infinitely many pairs of composite positive integers m and n such that $\sigma(m) = \varphi(n)$.

12. Do there exist positive integers x, y, z such that

$$\frac{x}{y} + \frac{y}{z} = \frac{z}{x} \quad ?$$

13. Find all positive integers n , $n > 1$, for which there exists an infinite sequence $a_1, a_2, a_3 \dots$ of nonzero integers such that for all positive integers k we have

$$a_k + 2a_{2k} + \dots + na_{nk} = 0.$$

14. Find all real values of the parameter a for which the equation

$$z^3 + az + 1 = 0$$

has a solution of the form $p + iq\sqrt{3}$ for some positive rationals p and q .

15. Let A be a subset of $\{1, 2, 3, \dots, 2n\}$ with more than n elements. Prove that there are two distinct elements of A whose sum is a prime number.
16. Let $\mathbb{P}$ be the set of all prime numbers. Find all functions $f : \mathbb{P} \rightarrow \mathbb{P}$ such that:

$$f(p)^{f(q)} + q^p = f(q)^{f(p)} + p^q$$

holds for all $p, q \in \mathbb{P}$.

17. Suppose that k is a fixed positive integer, and let n be the smallest positive integer with exactly k divisors. If n is a perfect cube, is it possible for k to have a prime divisor of the form $3j + 2$?